



Lecture 10

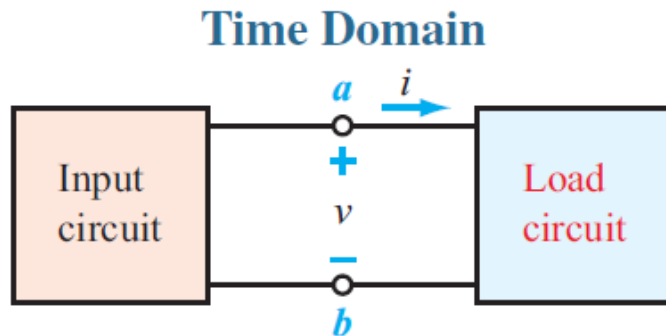
- AC Power Calculation



Outline

- Instantaneous power
- Average power
- Apparent power
- Power Factor
- Complex power

AC Power in Time Domain: Instantaneous



$$v(t) = V_m \cos(\omega t + \theta_v) \quad i(t) = I_m \cos(\omega t + \theta_i)$$

Instantaneous power:
power at any instant of time.

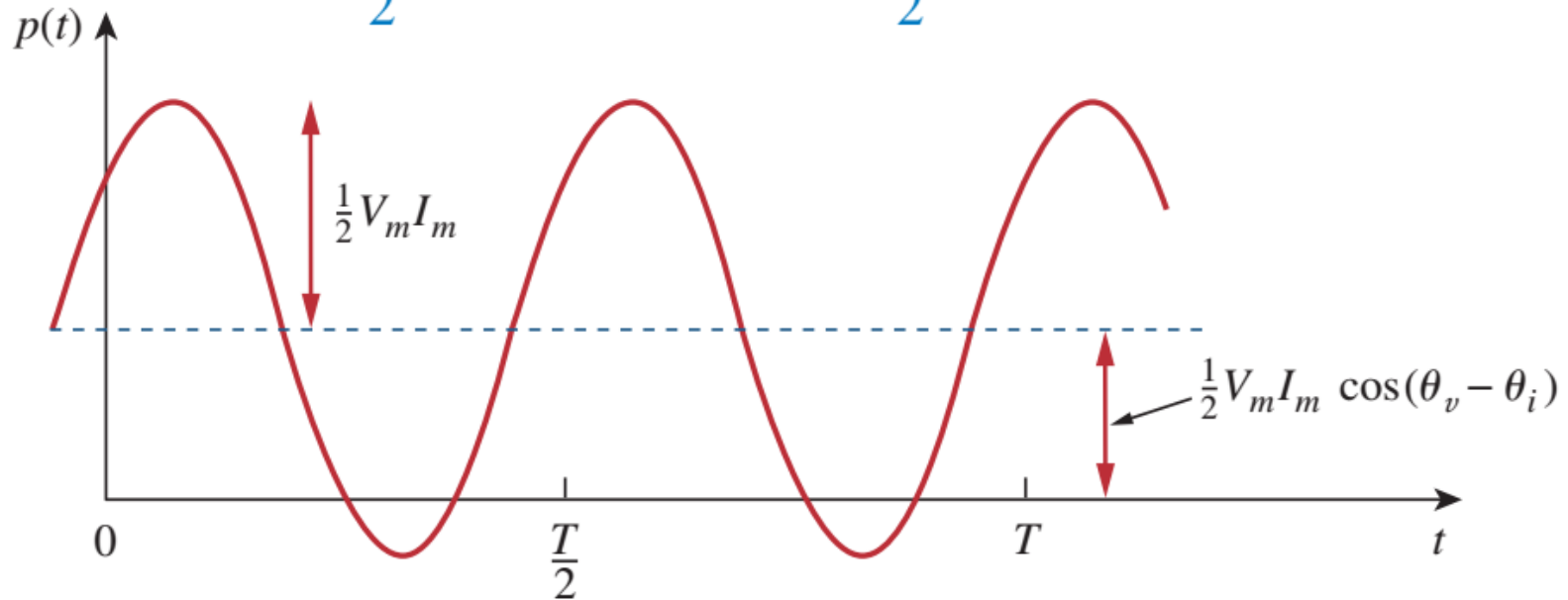
$$p(t) = v(t)i(t) = V_m I_m \cos(\omega t + \theta_v) \cos(\omega t + \theta_i)$$

$$p(t) = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) + \frac{1}{2} V_m I_m \cos(2\omega t + \theta_v + \theta_i)$$



AC Power in Time Domain: Instantaneous

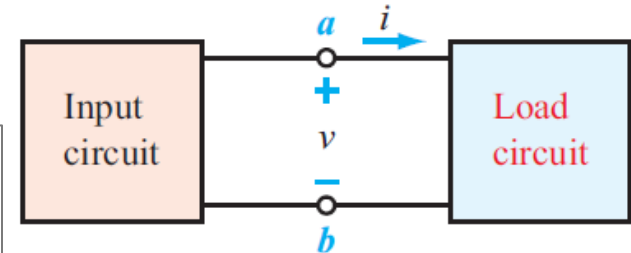
$$p(t) = \frac{1}{2}V_m I_m \cos(\theta_v - \theta_i) + \frac{1}{2}V_m I_m \cos(2\omega t + \theta_v + \theta_i)$$



Average Power P (*Capitalized*)

$$v(t) = V_m \cos(\omega t + \theta_v) \quad i(t) = I_m \cos(\omega t + \theta_i)$$

$$p(t) = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) + \frac{1}{2} V_m I_m \cos(2\omega t + \theta_v + \theta_i)$$



Average (or real) power (unit: watts)

The **average power**, in watts, is the average of the instantaneous power over one period.

$$P = \frac{1}{T} \int_0^T p(t) dt$$



Average Power P (time domain)

$$\begin{aligned} P &= \frac{1}{T} \int_0^T \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) dt \\ &\quad + \frac{1}{T} \int_0^T \frac{1}{2} V_m I_m \cos(2\omega t + \theta_v + \theta_i) dt \\ &= \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) \frac{1}{T} \int_0^T dt \\ &\quad + \frac{1}{2} V_m I_m \frac{1}{T} \int_0^T \cos(2\omega t + \theta_v + \theta_i) dt \end{aligned}$$

$$P = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i)$$



Average Power P (phasor domain)

$$v(t) = V_m \cos(\omega t + \theta_v) \quad i(t) = I_m \cos(\omega t + \theta_i)$$

$$\mathbf{V} = V_m \angle \theta_v \text{ and } \mathbf{I} = I_m \angle \theta_i,$$

$$\frac{1}{2} \mathbf{V} \mathbf{I}^* = \frac{1}{2} V_m I_m \angle \theta_v - \theta_i$$

$$= \frac{1}{2} V_m I_m [\cos(\theta_v - \theta_i) + j \sin(\theta_v - \theta_i)]$$

$$P = \frac{1}{2} \text{Re}[\mathbf{V} \mathbf{I}^*] = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i)$$



Two special cases for average power P

- For a purely resistive load R :

$$P = \frac{1}{2} V_m I_m = \frac{1}{2} I_m^2 R = \frac{1}{2} |\mathbf{I}|^2 R \quad \text{where } |\mathbf{I}|^2 = \mathbf{I} \times \mathbf{I}^*$$

- For a purely reactive load:

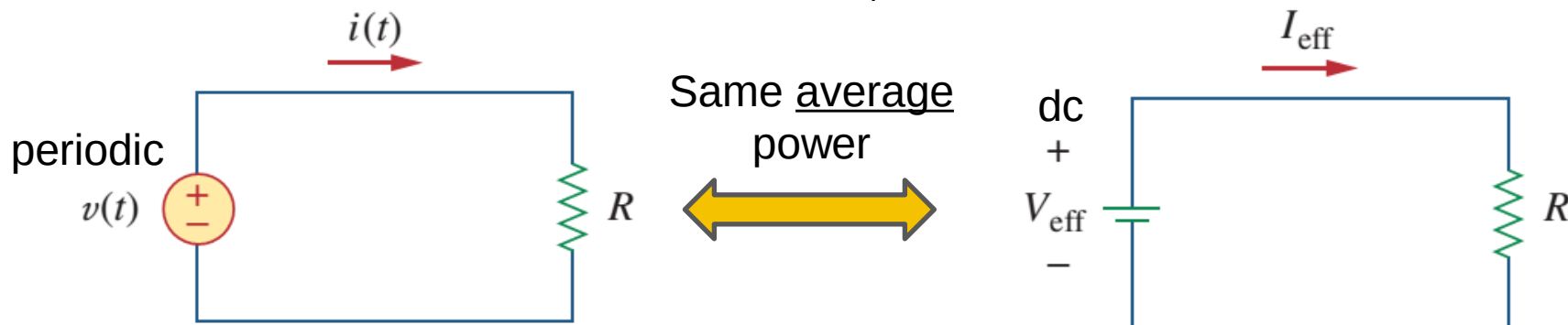
$$P = \frac{1}{2} V_m I_m \cos 90^\circ = 0$$

A resistive load (R) absorbs power at all times, while a reactive load (L or C) absorbs zero average power.

Effective Value (RMS)

- For any periodic function $x(t)$ in general, its rms value is

$$X_{\text{eff}} = X_{\text{rms}} = \sqrt{\frac{1}{T} \int_0^T x^2(t) dt}$$



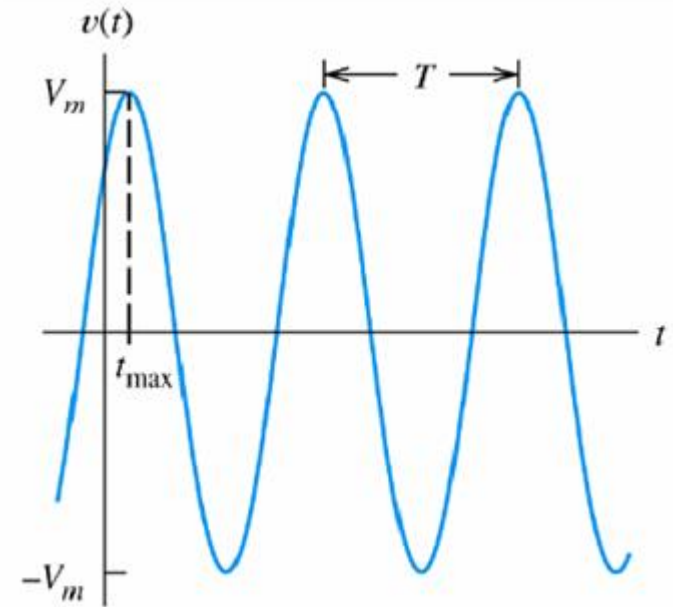
$$I_{\text{eff}} = \sqrt{\frac{1}{T} \int_0^T i^2(t) dt}$$

$$V_{\text{eff}} = \sqrt{\frac{1}{T} \int_0^T v^2(t) dt}$$

Example: RMS of a Sinusoidal

- The RMS value of $v(t) = V_m \cos(\omega t + \phi)$ is

$$\begin{aligned} V_{\text{rms}} &= \sqrt{\frac{1}{T} \int_{t_0}^{t_0+T} v^2(t) dt} \\ &= \sqrt{\frac{1}{T} \int_{t_0}^{t_0+T} V_m^2 \cdot \cos^2(\omega t + \phi) dt} \\ &= \frac{V_m}{\sqrt{2}} \end{aligned}$$



Average
Power

$$\begin{aligned} P &= \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) = \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} \cos(\theta_v - \theta_i) \\ &= V_{\text{rms}} I_{\text{rms}} \cos(\theta_v - \theta_i) \end{aligned}$$



Determine the rms value of the current waveform in Fig. 11.14. If the current is passed through a $2\text{-}\Omega$ resistor, find the average power absorbed by the resistor.

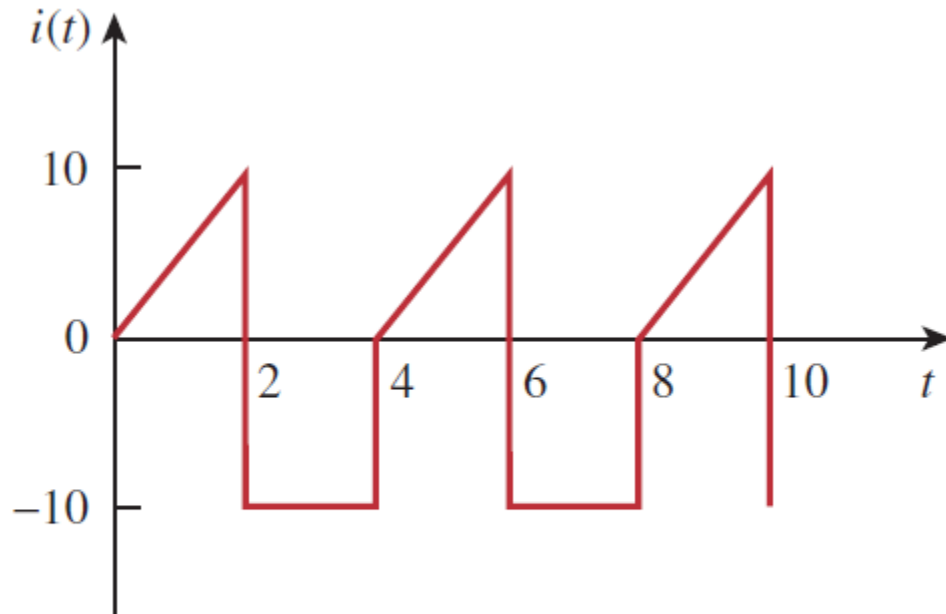


Figure 11.14
For Example 11.7.



The waveform shown in Fig. 11.16 is a half-wave rectified sine wave. Find the rms value and the amount of average power dissipated in a $10\text{-}\Omega$ resistor.

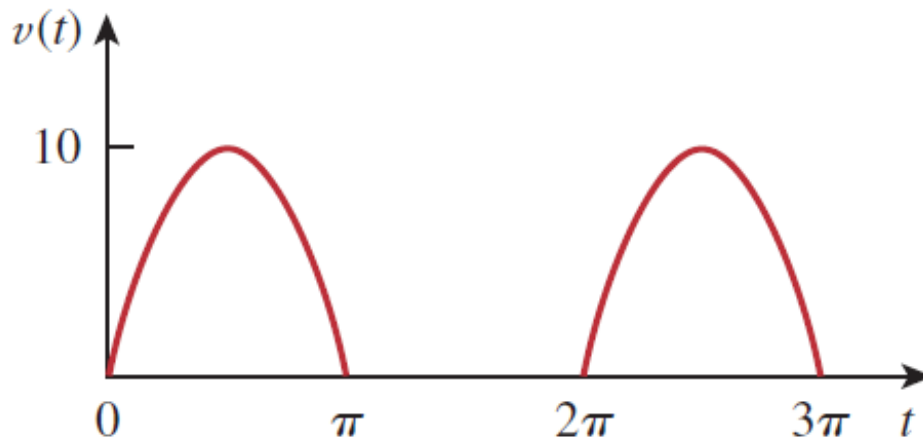


Figure 11.16
For Example 11.8.

Apparent Power

$$P = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) = \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} \cos(\theta_v - \theta_i) \\ = V_{\text{rms}} I_{\text{rms}} \cos(\theta_v - \theta_i)$$

$$S = V_{\text{rms}} I_{\text{rms}}$$

Unit: volt-amp (VA)

It seems apparent that the power should be the voltage-current product, by analogy with dc resistive circuits.

Power Factor

$$P = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) = \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} \cos(\theta_v - \theta_i) \\ = V_{\text{rms}} I_{\text{rms}} \cos(\theta_v - \theta_i)$$

- The power factor

$$\text{pf} = \frac{P}{S} = \cos(\theta_v - \theta_i)$$

- $(\theta_v - \theta_i)$ is called power factor angle.
 - >0 means a *lagging* pf (current lags voltage)
 - <0 means a *leading* pf (current leads voltage)

Power Factor-2

$$P = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) = \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} \cos(\theta_v - \theta_i) \\ = V_{\text{rms}} I_{\text{rms}} \cos(\theta_v - \theta_i)$$

- The power factor

$$\text{pf} = \frac{P}{S} = \cos(\theta_v - \theta_i)$$

- $(\theta_v - \theta_i)$ is called power factor angle.
- $(\theta_v - \theta_i)$ is equal to the angle of the load impedance

$$\mathbf{Z} = \frac{\mathbf{V}}{\mathbf{I}} = \frac{V_m \angle \theta_v}{I_m \angle \theta_i} = \frac{V_m}{I_m} \angle \theta_v - \theta_i$$

Also
$$\mathbf{Z} = \frac{\mathbf{V}}{\mathbf{I}} = \frac{V_{\text{rms}}}{I_{\text{rms}}} \angle \theta_v - \theta_i$$



Obtain the power factor and the apparent power of a load whose impedance is $\mathbf{Z} = 60 + j40 \, \Omega$ when the applied voltage is $v(t) = 320 \cos(377t + 10^\circ) \text{ V}$.

Determine the power factor of the entire circuit of Fig. 11.18 as seen by the source. Calculate the average power delivered by the source.

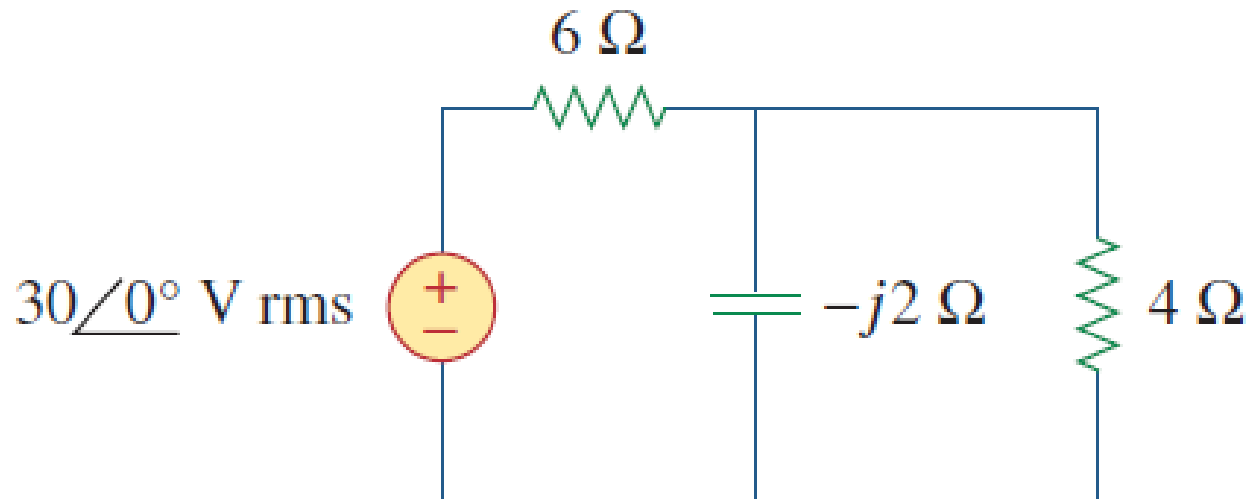


Figure 11.18

For Example 11.10.



Outline

- Instantaneous power
- Average power
- Apparent power
- Power Factor
- **Complex power**



Complex Power

$$v(t) = V_m \cos(\omega t + \theta_v) \Rightarrow \mathbf{V} = V_m \angle \theta_v$$

$$i(t) = I_m \cos(\omega t + \theta_i) \Rightarrow \mathbf{I} = I_m \angle \theta_i$$

We observe that:

$$\begin{aligned} \frac{1}{2} \mathbf{V} \mathbf{I}^* &= \frac{1}{2} V_m I_m \angle (\theta_v - \theta_i) \\ &= \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) + j \frac{1}{2} V_m I_m \sin(\theta_v - \theta_i) \end{aligned}$$

- Define a *single* power metric

$$\mathbf{S} = \frac{1}{2} \mathbf{V} \mathbf{I}^* = \mathbf{V}_{\text{rms}} \mathbf{I}_{\text{rms}}^* = V_{\text{rms}} I_{\text{rms}} \angle (\theta_v - \theta_i) = P + jQ$$

Unit: volt-amp (VA)

Complex power is important in power analysis because it contains *all* the information pertaining to the power absorbed by a given load.

Another Way to Calculate Complex Power

$$S = I_{\text{rms}}^2 Z = \frac{V_{\text{rms}}^2}{Z^*} = V_{\text{rms}} I_{\text{rms}}^*$$

$$S = V_{\text{rms}} I_{\text{rms}}^*$$

$$= V_{\text{rms}} \left(\frac{V_{\text{rms}}}{Z} \right)^*$$

$$= \frac{|V_{\text{rms}}|^2}{Z^*}$$

$$S = V_{\text{rms}} I_{\text{rms}}^*$$

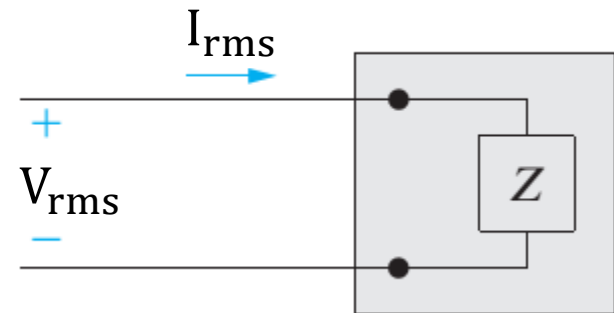
$$= I_{\text{rms}} Z I_{\text{rms}}^*$$

$$= |I_{\text{rms}}|^2 Z$$

$$= |I_{\text{rms}}|^2 (R + jX)$$

$$= |I_{\text{rms}}|^2 R + j |I_{\text{rms}}|^2 X$$

$$= I_{\text{rms}}^2 R + j I_{\text{rms}}^2 X$$



$$V_{\text{rms}} = I_{\text{rms}} Z$$

$$P = \text{Re}(S) = I_{\text{rms}}^2 R$$

$$Q = \text{Im}(S) = I_{\text{rms}}^2 X$$



$$\mathbf{S} = \frac{1}{2} \mathbf{V} \mathbf{I}^* = \mathbf{V}_{\text{rms}} \mathbf{I}_{\text{rms}}^* = V_{\text{rms}} I_{\text{rms}} \angle(\theta_v - \theta_i) = P + jQ$$

$$P = \text{Re}(\mathbf{S}) = I_{\text{rms}}^2 R$$

$$Q = \text{Im}(\mathbf{S}) = I_{\text{rms}}^2 X$$

- Average (or real) power

$$P = \text{Re}\left[\frac{1}{2} \mathbf{V} \mathbf{I}^*\right]$$

Unit: Watts

- Reactive power

$$Q = \text{Im}\left[\frac{1}{2} \mathbf{V} \mathbf{I}^*\right]$$

Unit: Volt Amperes Reactive (VARs)

- Apparent power

$$s = |\mathbf{S}| = \frac{1}{2} V_m I_m = V_{\text{rms}} I_{\text{rms}}$$

Unit: volt-amp (VA)



$$\text{Complex Power} = \mathbf{S} = P + jQ = \mathbf{V}_{\text{rms}}(\mathbf{I}_{\text{rms}})^*$$

$$= |\mathbf{V}_{\text{rms}}| |\mathbf{I}_{\text{rms}}| \angle \theta_v - \theta_i$$

$$\text{Apparent Power} = S = |\mathbf{S}| = |\mathbf{V}_{\text{rms}}| |\mathbf{I}_{\text{rms}}| = \sqrt{P^2 + Q^2}$$

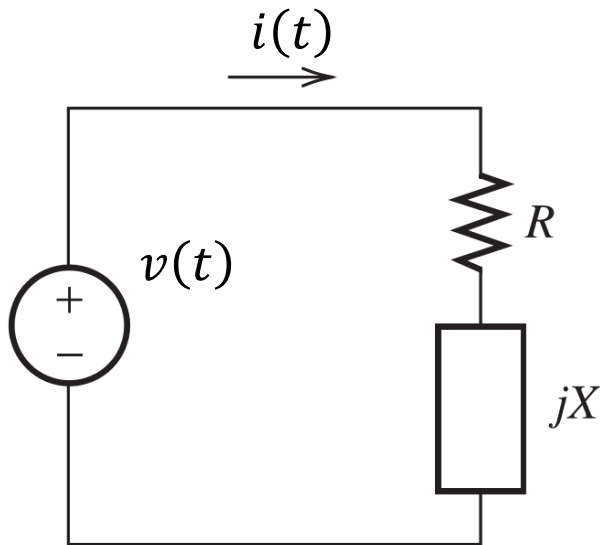
$$\text{Real Power} = P = \text{Re}(\mathbf{S}) = S \cos(\theta_v - \theta_i)$$

$$\text{Reactive Power} = Q = \text{Im}(\mathbf{S}) = S \sin(\theta_v - \theta_i)$$

$$\text{Power Factor} = \frac{P}{S} = \cos(\theta_v - \theta_i)$$

Example

- Find the average power and reactive power absorbed by an impedance $Z = 30 - j70\Omega$, when a voltage $V = 120\angle 0^\circ$ is applied across it.



$$\mathbf{I} = \frac{\mathbf{V}}{\mathbf{Z}} = \frac{120\angle 0^\circ}{30 - j70} = \frac{120\angle 0^\circ}{76.16\angle -66.8^\circ} \\ = 1.576\angle 66.8^\circ \text{ A}$$

$$P = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) = 37.24 \text{ W}$$

$$Q = \frac{1}{2} V_m I_m \sin(\theta_v - \theta_i) = -86.91 \text{ VAR}$$

Power Triangle

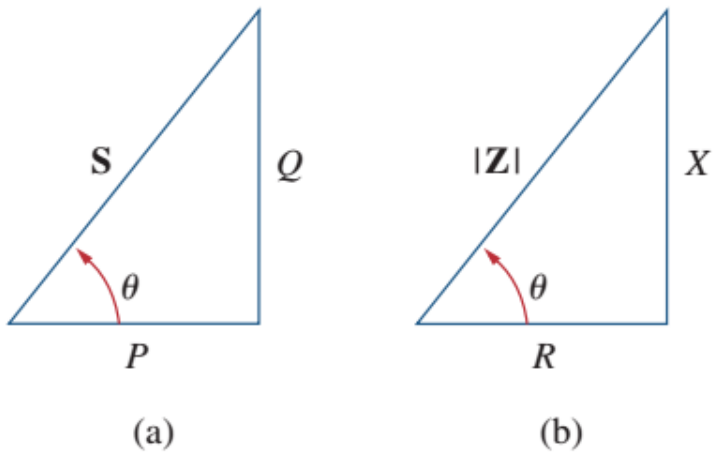


Figure 11.21

(a) Power triangle, (b) impedance triangle.

Quantity	Units
Complex power	volt-amps
Average power	watts
Reactive power	var

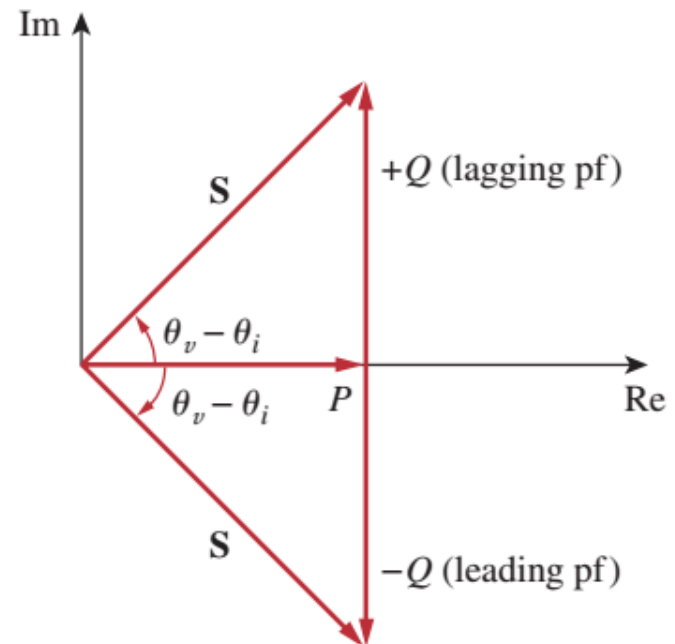
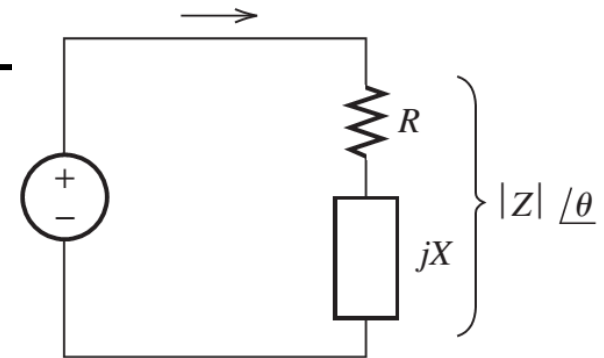


Figure 11.22

Power triangle.

Power Factor



Power factor leading and lagging relationships for a load $\mathbf{Z} = R + jX$.

Load Type	$\phi_z = \phi_v - \phi_i$	I-V Relationship	pf
Purely Resistive ($X = 0$)	$\phi_z = 0$	$\mathbf{I}$ in-phase with $\mathbf{V}$	1
Inductive ($X > 0$)	$0 < \phi_z \leq 90^\circ$	$\mathbf{I}$ lags $\mathbf{V}$	lagging
Purely Inductive ($X > 0$ and $R = 0$)	$\phi_z = 90^\circ$	$\mathbf{I}$ lags $\mathbf{V}$ by 90°	lagging
Capacitive ($X < 0$)	$-90^\circ \leq \phi_z < 0$	$\mathbf{I}$ leads $\mathbf{V}$	leading
Purely Capacitive ($X < 0$ and $R = 0$)	$\phi_z = -90^\circ$	$\mathbf{I}$ leads $\mathbf{V}$ by 90°	leading

Example

- A series-connected load draws a current

$$i(t) = 4\cos(100\pi t + 10^\circ)\text{A}$$

when the applied voltage is

$$v(t) = 120\cos(100\pi t - 20^\circ)\text{V}$$

- Find the apparent power and the power factor of the load.
- Determine the values that form the series-connected load.

$$V_{\text{rms}} I_{\text{rms}} = 240 \text{ VA}$$

$$\text{pf} = \cos(\theta_v - \theta_i) = 0.866 \quad (\text{leading})$$

$$\mathbf{Z} = \frac{\mathbf{V}}{\mathbf{I}} = 25.98 - j15 \, \Omega$$

$$C = \frac{1}{15\omega} = \frac{1}{15 \times 100\pi} = 212.2 \, \mu\text{F}$$

Exercise

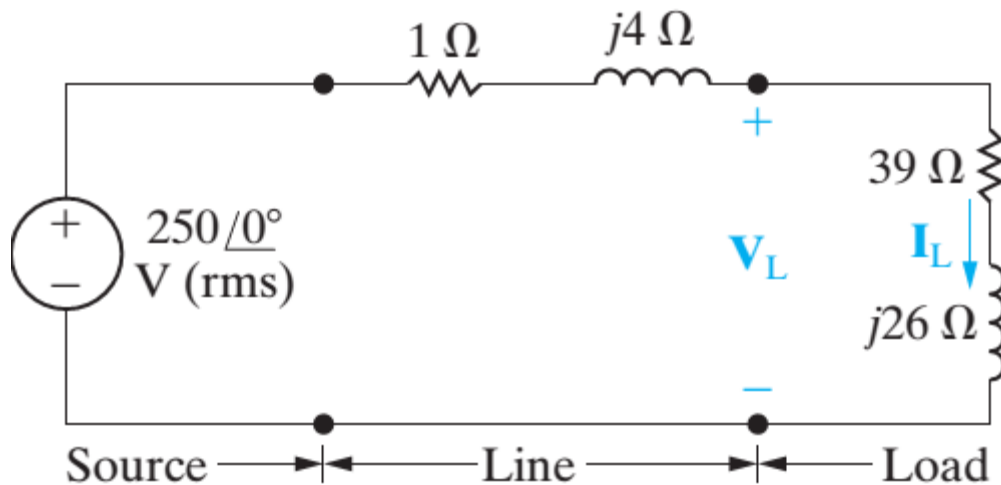
- The voltage across a load is $v(t) = 60\cos(\omega t - 10^\circ)\text{V}$, and the current through the load is $i(t) = 1.5\cos(\omega t + 50^\circ)$. Find
 - The complex and apparent powers.
 - The real and reactive powers.
 - The power factor and the load impedance.

$$\mathbf{S} = \mathbf{V}_{\text{rms}} \mathbf{I}_{\text{rms}}^* = 45 \angle -60^\circ \text{ VA}$$

$$\text{pf} = 0.5 \text{ (leading)}$$

$$\mathbf{Z} = \frac{\mathbf{V}}{\mathbf{I}} = 40 \angle -60^\circ \Omega$$

Example



- Find V_L and I_L .
- Find the average and reactive power
 - Delivered to the load
 - Delivered to the line
 - Supplied by the source

$$\begin{aligned} I_L &= \frac{250\angle 0^\circ}{40 + j30} = 4 - j3 \\ &= 5\angle -36.87^\circ \text{ (rms)} \end{aligned}$$

$$\begin{aligned} V_L &= I_L(39 + j26) \\ &= 234 - j13 \\ &= 234.36\angle -3.18^\circ \end{aligned}$$

Load:

$$V_L I_L^* = 975 + j650 \text{ VA}$$

Line:

$$P = (5)^2(1) = 25 \text{ W}$$

$$Q = (5)^2(4) = 100 \text{ VAR}$$

Source:

$$250\angle 0^\circ I_L^* = 1000 + j750 \text{ VA}$$



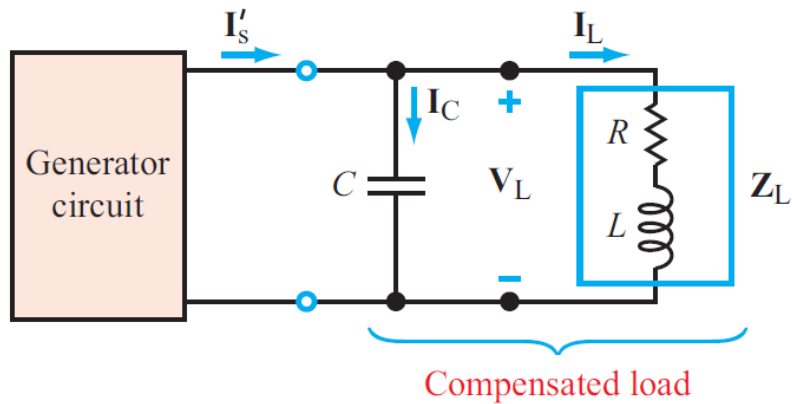
Conservation of AC Power

Practice Problem 11.14

Two loads connected in parallel are respectively 2 kW at a pf of 0.75 leading and 4 kW at a pf of 0.95 lagging. Calculate the pf of the two loads. Find the complex power supplied by the source.

Answer: 0.9972 (leading), $6 - j0.4495$ kVA.

- Power factor correction



- Maximum power transfer

